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Monopoly

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Why Monopolies Arise

- Monopoly
 - Firm that is the sole seller of a product without close substitutes
 - Price maker
 - Charges a price that exceeds marginal cost



Barriers to entry

- Monopoly resources
- Government regulation
- Natural monopoly

Table 1

A Monopoly's Total, Average, and Marginal Revenue

Quantity of Water (Q)	Price (P)	Total Revenue (TR = P x Q)	Average Revenue (AR = TR/Q)	Marginal Revenue (MR = $\Delta TR / \Delta Q$)
0 gallons	\$11	\$0		
1	10	10	\$10	\$10
2	9	18	9	8
3	8	24	8	6
4	7	28	7	4
5	6	30	6	2
6	5	30	5	0
7	4	28	4	-2
8	3	24	3	-4



Production and Pricing Decisions

- Monopoly
 - Price maker
 - Sole producer
 - Downward sloping demand: the market demand curve
- Competitive firm
 - Price taker
 - One producer of many
 - Demand is a horizontal line (Price)

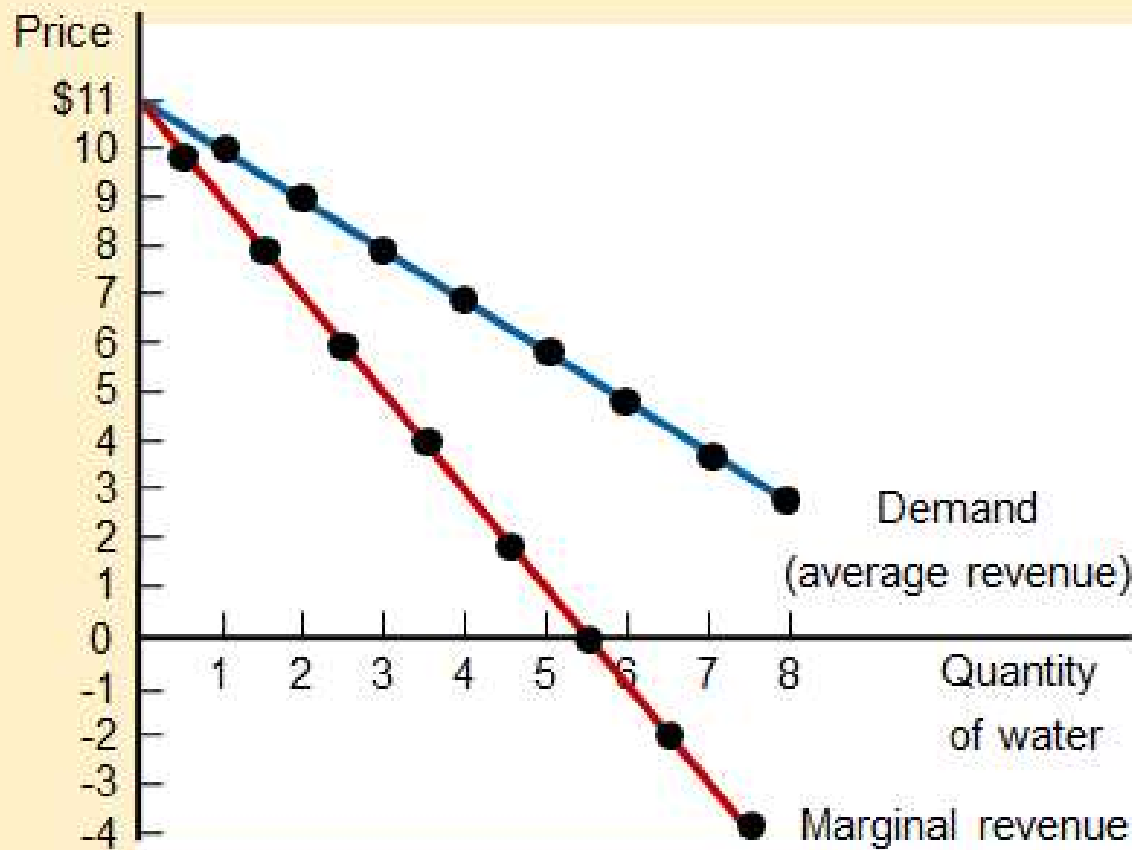


Production and Pricing Decisions

- Increase in quantity sold
 - Output effect
 - Q is higher: increase total revenue
 - Price effect
 - P is lower: decrease total revenue
- Because $MR < P$
 - Marginal-revenue curve is below the demand curve

Figure 3

Demand and Marginal-Revenue Curves for a Monopoly



The demand curve shows how the quantity affects the price of the good. The marginal-revenue curve shows how the firm's revenue changes when the quantity increases by 1 unit. Because the price on *all* units sold must fall if the monopoly increases production, marginal revenue is always less than the price.

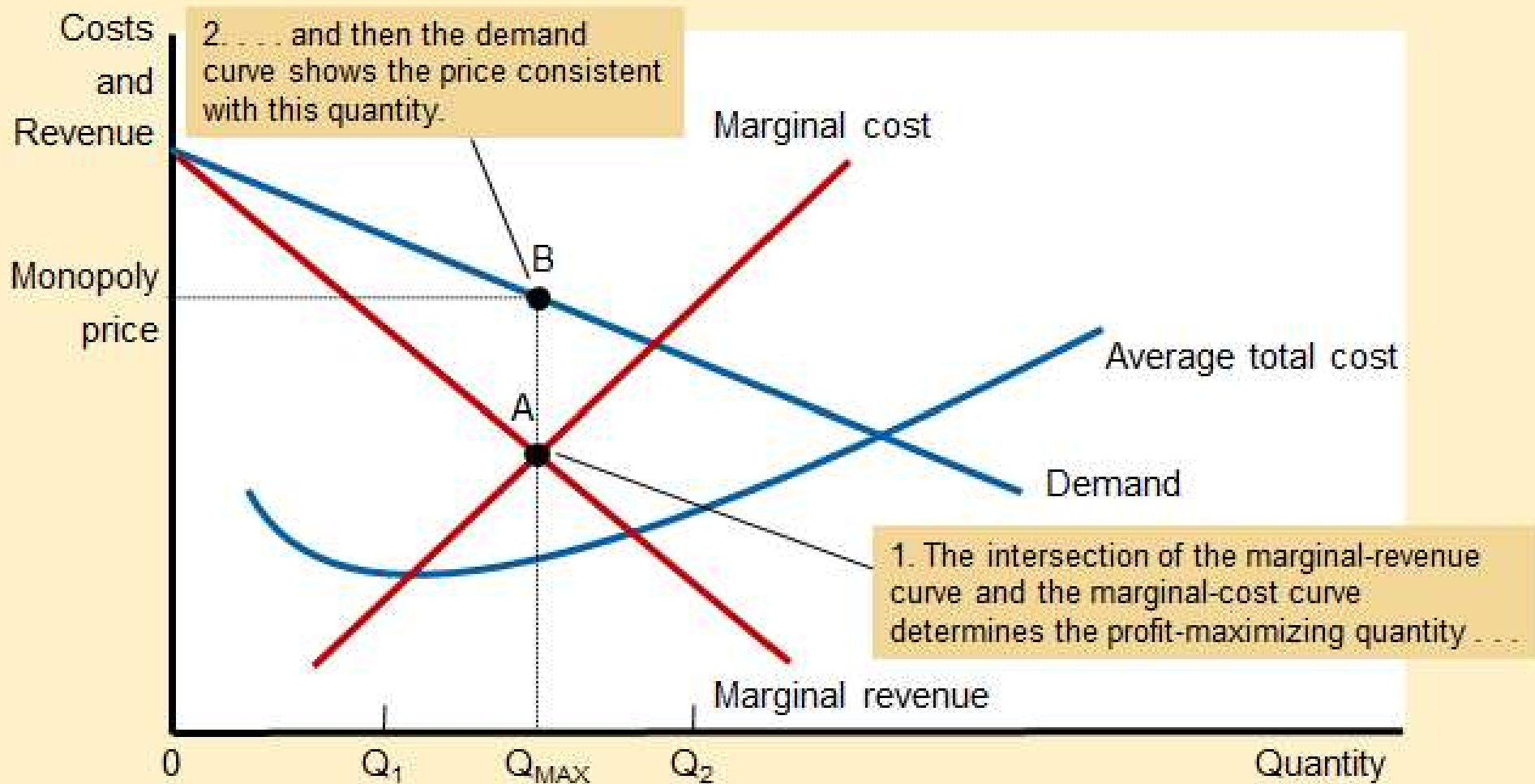


Production and Pricing Decisions

- Profit maximization
 - If $MR > MC$: increase production
 - If $MC > MR$: produce less
 - Maximize profit
 - Produce quantity where $MR=MC$
 - Intersection of the marginal-revenue curve and the marginal-cost curve
 - Price: on the demand curve

Figure 4

Profit Maximization for a Monopoly



A monopoly maximizes profit by choosing the quantity at which marginal revenue equals marginal cost (point A). It then uses the demand curve to find the price that will induce consumers to buy that quantity (point B).

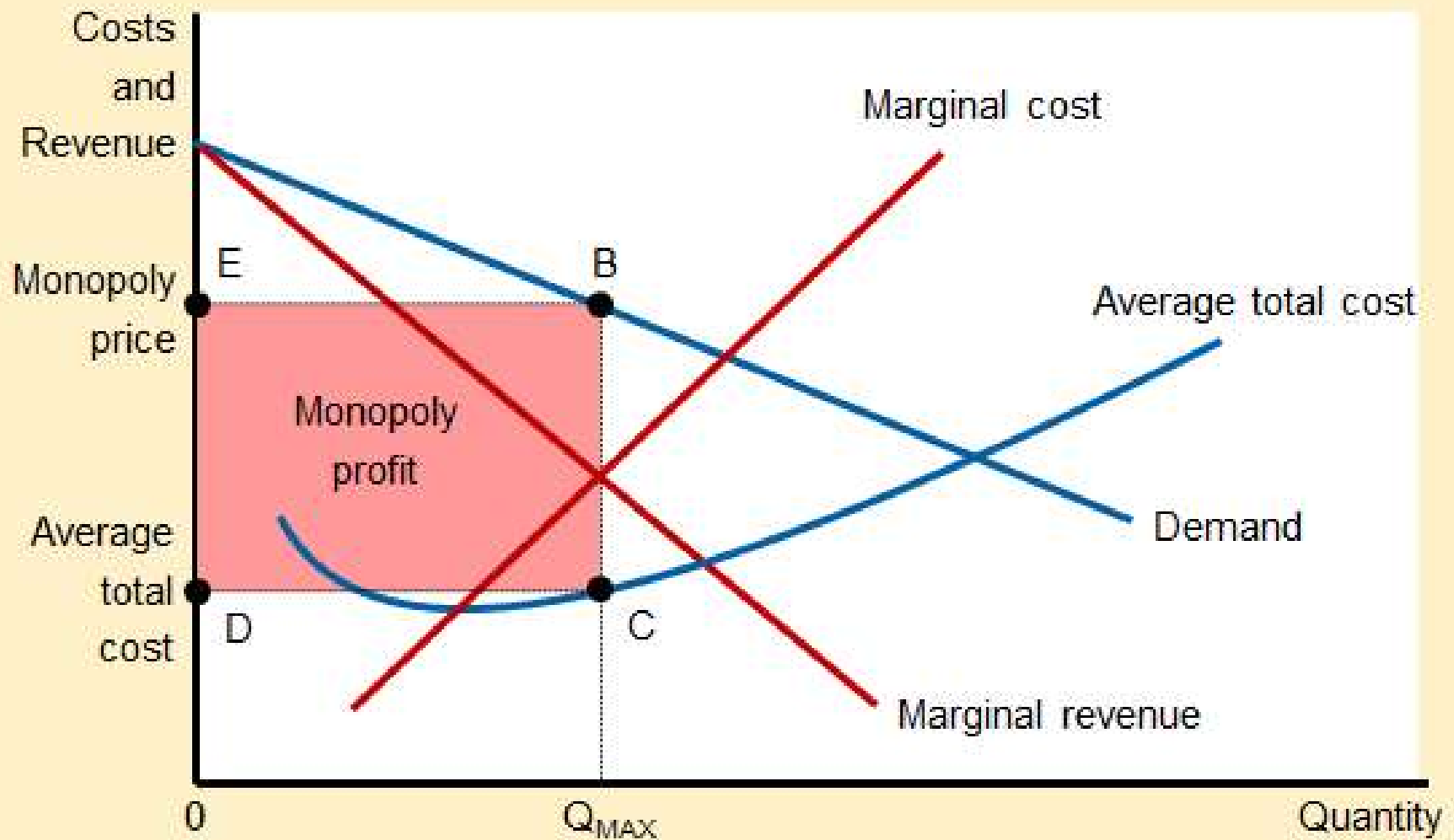


Production and Pricing Decisions

- Profit maximization
 - Perfect competition: $P=MR=MC$
 - Price equals marginal cost
 - Monopoly: $P>MR=MC$
 - Price exceeds marginal cost
- A monopoly's profit
 - Profit = $TR - TC = (P - ATC) \times Q$

Figure 5

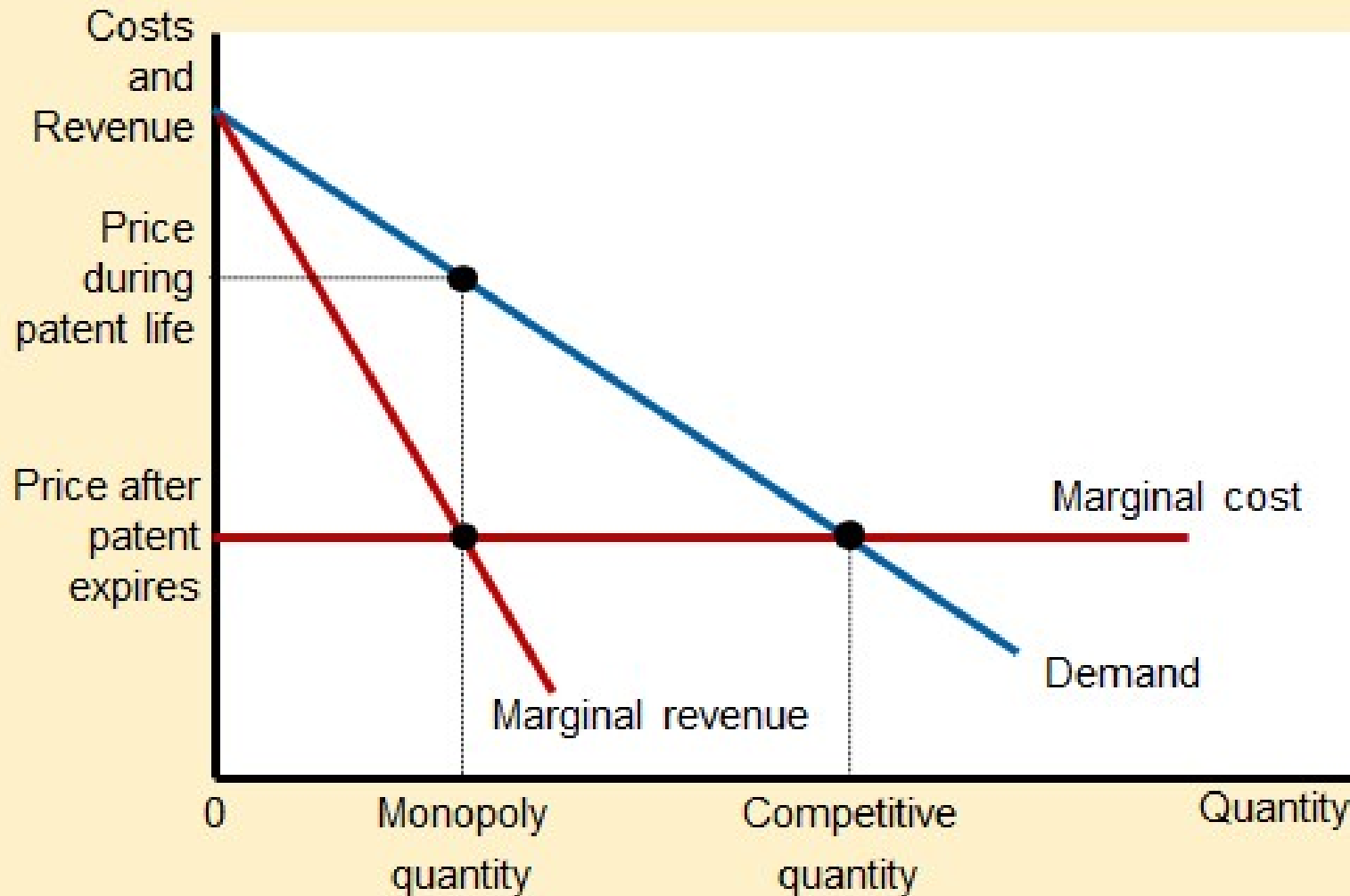
The Monopolist's Profit



The area of the box BCDE equals the profit of the monopoly firm. The height of the box (BC) is price minus average total cost, which equals profit per unit sold. The width of the box (DC) is the number of units sold.

Figure 6

Monopoly to perfect competition



When a patent gives a firm a monopoly over the sale of a drug, the firm charges the monopoly price, which is well above the marginal cost of making the drug. When the patent on a drug runs out, new firms enter the market, making it more competitive. As a result, the price falls from the monopoly price to marginal cost.



The Welfare Cost of Monopolies

- Total surplus
 - Economic well-being of buyers and sellers in a market
 - Sum of consumer surplus and producer surplus
- Consumer surplus
 - Consumers' willingness to pay for a good
 - Minus the amount they actually pay for it

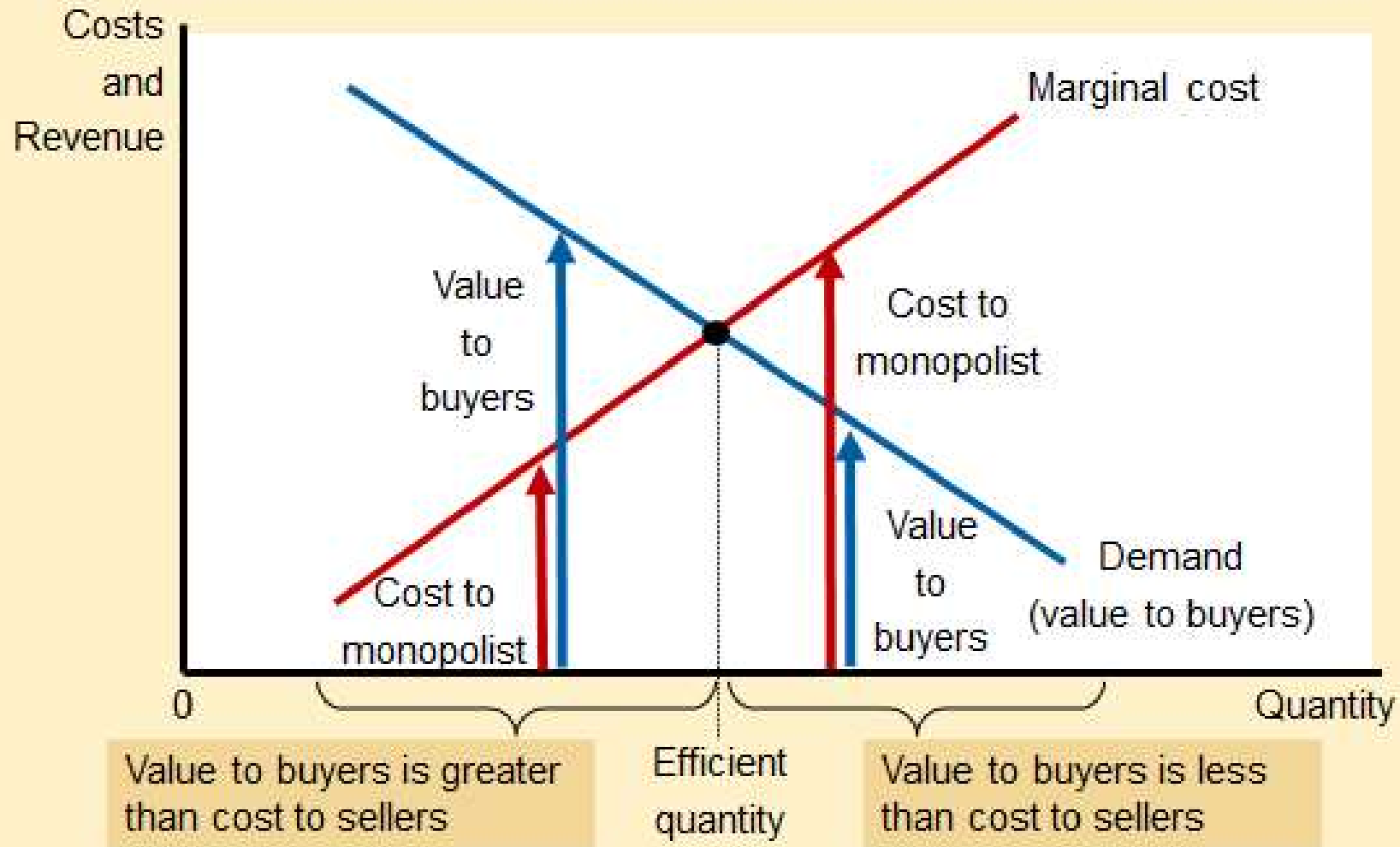


The Welfare Cost of Monopolies

- **Producer surplus**
 - Amount producers receive for a good
 - Minus their costs of producing it
- **Benevolent planner: maximize total surplus**
 - Socially efficient outcome
 - Produce quantity where
 - Marginal cost curve intersects demand curve
 - Charge $P=MC$

Figure 7

The Efficient Level of Output



A benevolent social planner maximizes total surplus in the market by choosing the level of output where the demand curve and marginal-cost curve intersect. Below this level, the value of the good to the marginal buyer (as reflected in the demand curve) exceeds the marginal cost of making the good. Above this level, the value to the marginal buyer is less than marginal cost.

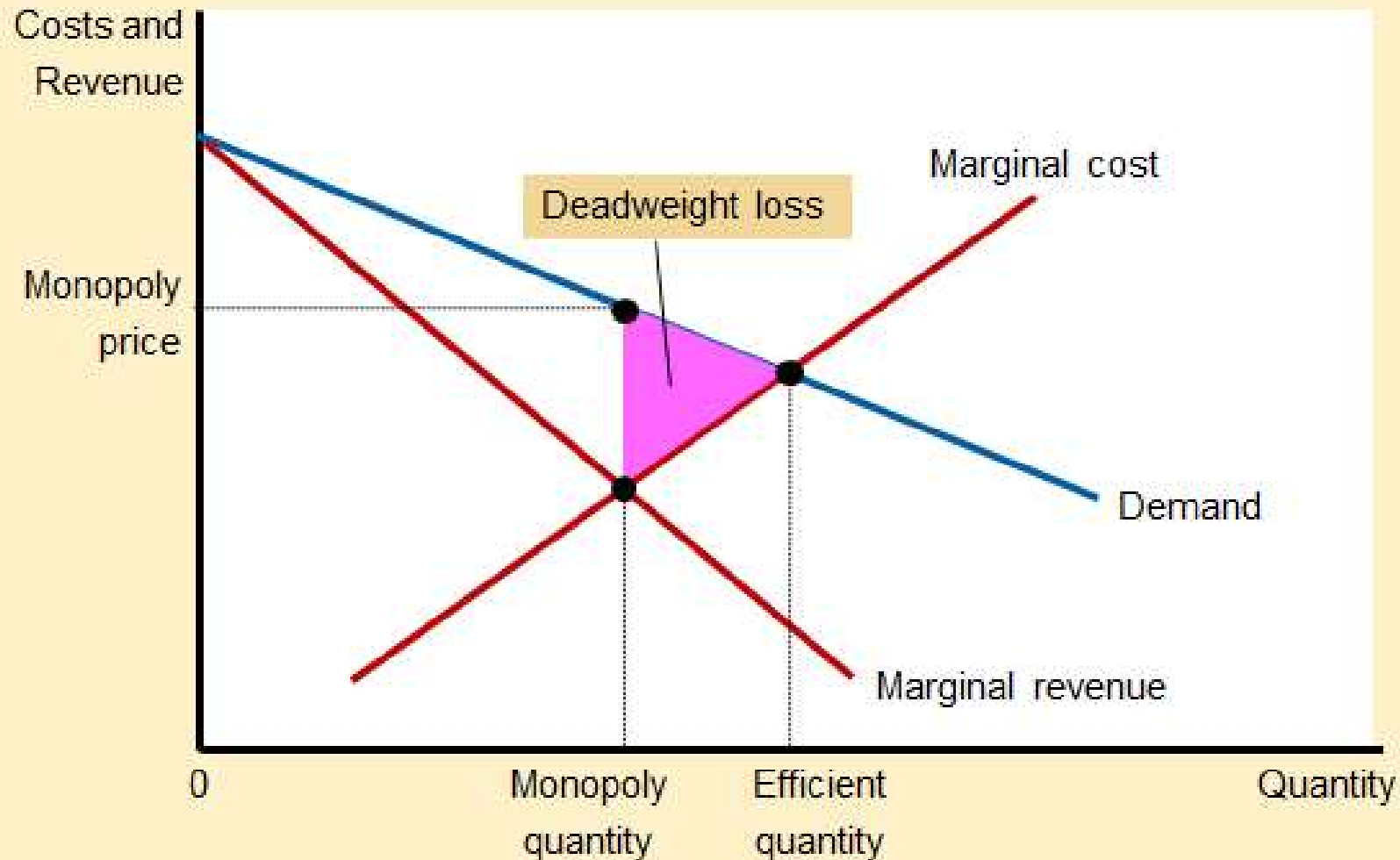


The Welfare Cost of Monopolies

- Monopoly
 - Produce quantity where $MC = MR$
 - Produces less than the socially efficient quantity of output
 - Charge $P > MC$
 - Deadweight loss
 - Triangle between the demand curve and MC curve

Figure 8

The Inefficiency of Monopoly



Because a monopoly charges a price above marginal cost, not all consumers who value the good at more than its cost buy it. Thus, the quantity produced and sold by a monopoly is below the socially efficient level. The deadweight loss is represented by the area of the triangle between the demand curve (which reflects the value of the good to consumers) and the marginal-cost curve (which reflects the costs of the monopoly producer).



The Welfare Cost of Monopolies

- The monopoly's profit: a social cost?
 - Monopoly - higher profit
 - Not a reduction of economic welfare
 - Bigger producer surplus
 - Smaller consumer surplus
 - Not a social problem
 - Social loss = Deadweight loss
 - From the inefficiently low quantity of output



Price Discrimination

- Price discrimination
 - Business practice
 - Sell the same good at different prices to different customers
 - Rational strategy to increase profit
 - Requires the ability to separate customers according to their willingness to pay
 - Can raise economic welfare



Price Discrimination

- Perfect price discrimination
 - Charge each customer a different price
 - Exactly his or her willingness to pay
 - Monopoly firm gets the entire surplus (Profit)
 - No deadweight loss

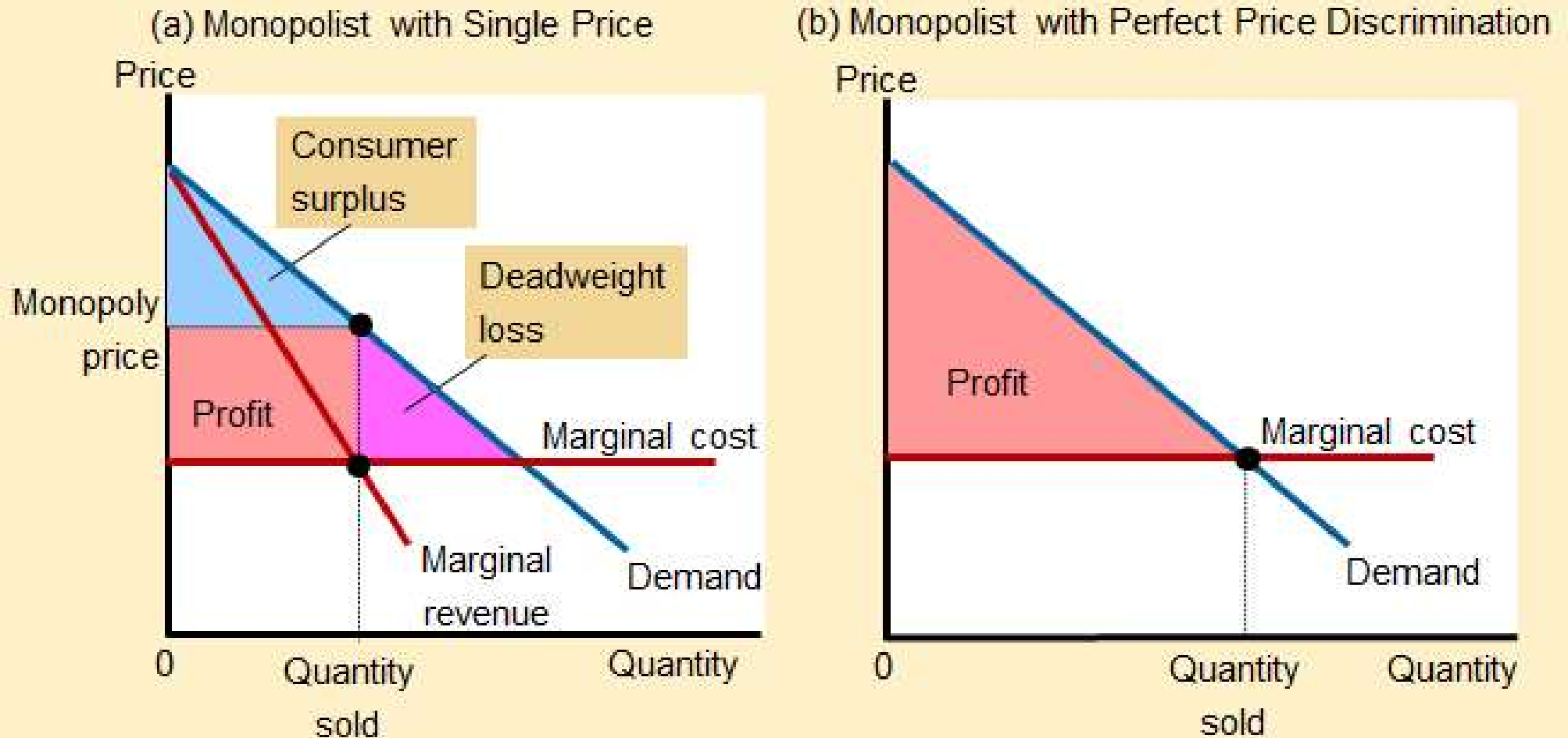


Price Discrimination

- Without price discrimination
 - Single price $>$ MC
 - Consumer surplus
 - Producer surplus (Profit)
 - Deadweight loss

Figure 9

Welfare with and without Price Discrimination



Panel (a) shows a monopoly that charges the same price to all customers. Total surplus in this market equals the sum of profit (producer surplus) and consumer surplus. Panel (b) shows a monopoly that can perfectly price discriminate. Because consumer surplus equals zero, total surplus now equals the firm's profit. Comparing these two panels, you can see that perfect price discrimination raises profit, raises total surplus, and lowers consumer surplus.



Price Discrimination

- Examples of price discrimination
 - Movie tickets
 - Lower price for children and seniors
 - Airline prices
 - Lower price for round-trip with Saturday night stay

"Would it bother you to hear how little I paid for this flight?"





Price Discrimination

- Examples of price discrimination
 - Discount coupons
 - Not all customers are willing to spend time to clip coupons
 - Financial aid
 - High tuition and need-based financial aid
 - Willingness to pay
 - Quantity discounts
 - Customer pays a higher price for the first unit bought than for the last unit bought



Public Policy Toward Monopolies

- Increasing competition with antitrust laws
 - Sherman Antitrust Act, 1890
 - Clayton Antitrust Act, 1914
 - Prevent mergers
 - Break up companies
 - Prevent companies from coordinating their activities to make markets less competitive



"But if we do merge with Amalgamated, we'll have enough resources to fight the anti-trust violation caused by the merger."



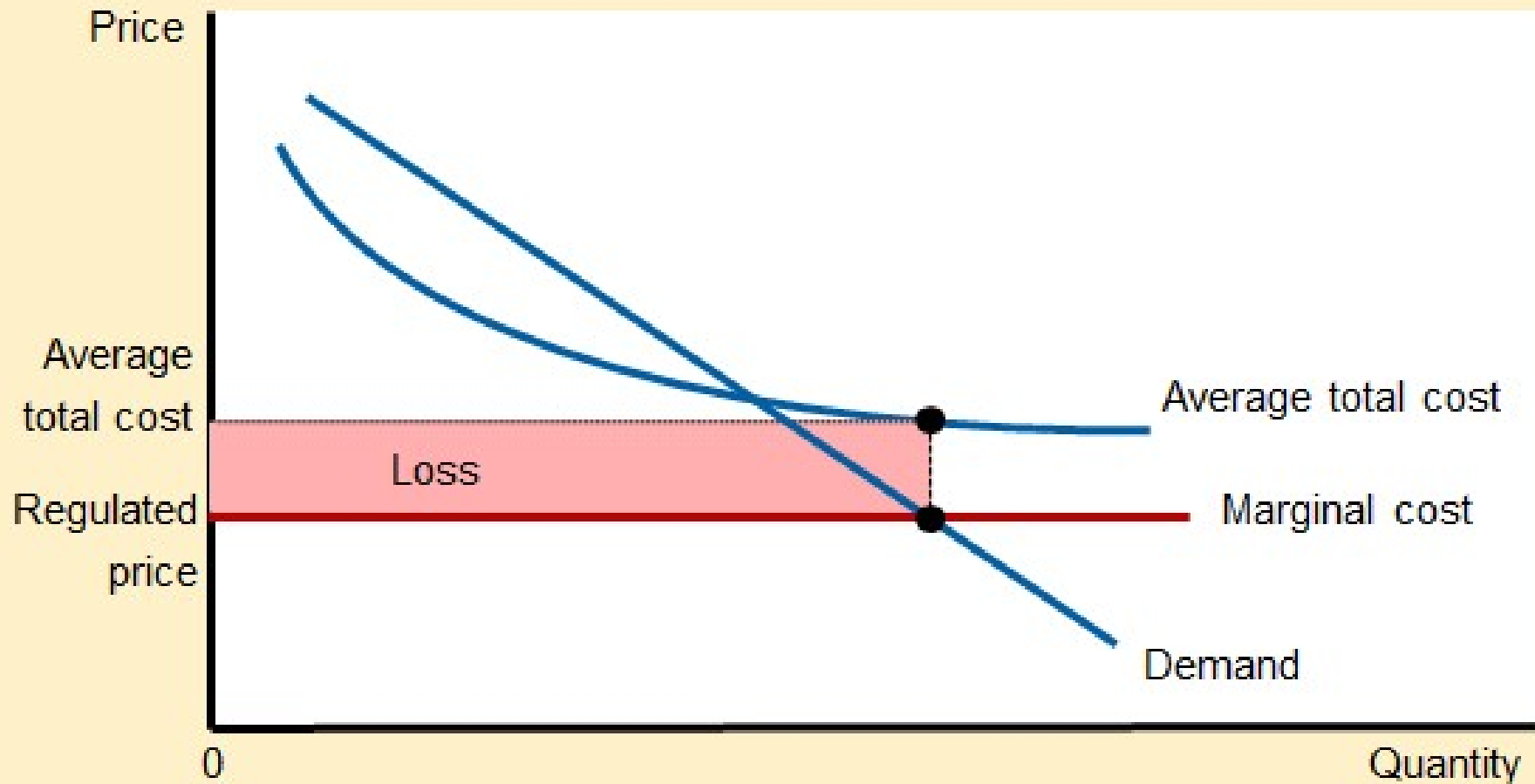
Public Policy Toward Monopolies

- Regulation

- Regulate the behavior of monopolists
 - Price
- Common in case of natural monopolies
- Marginal-cost pricing
 - May be less than ATC
 - No incentive to reduce costs

Figure 10

Marginal-Cost Pricing for a Natural Monopoly



Because a natural monopoly has declining average total cost, marginal cost is less than average total cost. Therefore, if regulators require a natural monopoly to charge a price equal to marginal cost, price will be below average total cost, and the monopoly will lose money.



Public Policy Toward Monopolies

- Public ownership
 - How the ownership of the firm affects the costs of production
 - Private owners
 - Incentive to minimize costs
 - Public owners (government)
 - If it does a bad job
 - Losers are the customers and taxpayers

Table 2

Competition versus Monopoly: A Summary Comparison

	Competition	Monopoly
Similarities		
Goal of firms	Maximize profits	Maximize profits
Rule for maximizing	$MR = MC$	$MR = MC$
Can earn economic profits in the short run?	Yes	Yes
Differences		
Number of firms	Many	One
Marginal revenue	$MR = P$	$MR < P$
Price	$P = MC$	$P > MC$
Produces welfare-maximizing level of output?	Yes	No
Entry in long run?	Yes	No
Can earn economic profits in long run?	No	Yes
Price discrimination possible?	No	Yes